

Tushar Vargude

+91 9359824101 | tvargude@gmail.com | LinkedIn | GitHub

Professional Summary

Aspiring AI & Data Science Engineer with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Generative AI. Skilled in data preprocessing, model development, testing, deployment, and AI-powered application development. Interested in Agentic AI, Large Language Models (LLMs), intelligent automation, workflow orchestration, and building innovative data-driven solutions.

Technical Skills

Programming	Python, SQL
AI/ML	Machine Learning, Deep Learning, Classification, Regression, Clustering, Model Training
Generative AI	LLMs, Prompt Engineering, Agentic AI, RAG, Intelligent Automation
Libraries	Pandas, NumPy, Scikit-learn, TensorFlow, OpenCV
Tools	Git, GitHub, Jupyter Notebook, Streamlit, Firebase
Concepts	Data Cleaning, Data Preprocessing, Feature Engineering, Workflow Orchestration, System Integration

Projects

Plant Disease Detection System 2025

- Developed a deep learning-based web application for plant disease classification.
- Performed image preprocessing, model training, and performance evaluation.
- Integrated Gemini API to generate AI-powered disease descriptions and recommendations.
- Technologies: Python, TensorFlow, OpenCV, Streamlit, Gemini API.

Traffic Density Detection System 2025

- Built a real-time vehicle detection and density estimation system using Computer Vision.
- Applied image preprocessing and feature extraction techniques.
- Simulated dynamic traffic signal timing based on traffic density.
- Technologies: Python, OpenCV, NumPy, Machine Learning.

Education

Sanjivani University 2024 – Present
B.Tech in Computer Science and Engineering (Artificial Intelligence & Data Science)

Sanjivani K.B.P Polytechnic, Kopergaon 2021 – 2024
Diploma in Mechatronics Engineering

Relevant Coursework

Machine Learning • Artificial Intelligence • Computer Vision • Database Management Systems • Data Structures & Algorithms • Deep Learning • Probability & Statistics

Languages

English (Fluent) • Hindi (Fluent) • Marathi (Native)